

SIGN: Saliency-Aware Integrated Global-Local Network for Cross-View Geo-Localization

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Abstract—Cross-View Geo-Localization (CVGL) matches images from different platforms, such as UAVs and satellites, to estimate target locations using geographic data. This task poses significant challenges due to viewpoint differences and scale inconsistencies. We propose a novel CVGL method named saliency-aware integrated global-local network (SIGN) to effectively address the viewpoint differences and scale inconsistencies between UAV and satellite images. SIGN employs a transformer-based network for fine-grained feature extraction and utilizes the Saliency-Aware Partition (SAP) module to divide images into key regions for local alignment. The model integrates global feature contrast learning with saliency-aware local block contrast learning to enhance alignment accuracy at both global and local levels. Experiments on the University-1652 dataset show that SIGN achieves state-of-the-art performance, demonstrating superior accuracy and robustness in handling viewpoint variations.

Index Terms—Cross-View Geo-Localization, Image Retrieval, Coarse-to-Fine, Deep Learning, Transformer

I. INTRODUCTION

Cross-View Geo-Localization (CVGL) tasks aim to match images from one viewpoint with the most similar image of the same geographic target from another viewpoint, which can be understood as a cross-platform image retrieval task. Traditional GPS-based localization methods often suffer from inaccuracies in complex geographic environments, such as mountainous areas, forests, or densely built urban areas, due to potential signal occlusion. As a result, CVGL has emerged as a promising alternative solution, gradually attracting the attention of researchers. Initially, CVGL research focused primarily on matching ground images with satellite images [1, 2]. However, the near-vertical viewpoints and significant differences in imaging modalities between these two platforms present considerable challenges for CVGL tasks. In recent years, with the rapid development of low-altitude Unmanned Aerial Vehicle (UAV) technology, UAV-based CVGL tasks have received increasing attention. Due to its low-altitude vantage points, UAVs are crucial in applications such as target tracking [3, 4], aerial surveillance [5, 6], and disaster response [7]. Compared to traditional ground images, UAV images can reduce interference from ground obstacles and provide more comprehensive and accurate location information. Zheng et al. [8] first introduced oblique-view UAV images into CVGL tasks, further expanding the application scope of CVGL, particularly making significant progress in leveraging

the advantages of UAV perspective images. Currently, CVGL applications can be broadly classified into two categories: UAV-based target localization and UAV navigation.

With the continuous development of deep learning [9, 10], Convolutional Neural Networks (CNNs) have been widely applied to various visual tasks, such as image classification [11], object detection [12], and semantic segmentation [13]. Many geo-localization methods utilize CNNs to extract image features and estimate the UAV's location by matching the visual features between UAV and satellite images [14–17]. However, CNNs have limitations in capturing global contextual information, which may result in insufficient modeling of global relationships in CVGL tasks, thus affecting localization accuracy.

In recent years, the transformer [18] have achieved remarkable success in various domains due to their superior ability to model global context. Unlike CNNs, transformers leverage self-attention mechanisms to directly capture long-range information, making them particularly well-suited for tasks that require global context understanding. In the field of computer vision (CV), the Vision Transformer (ViT) [19] has further demonstrated the power of transformer-based architectures. As a result, transformer-based CVGL methods have gradually become a research hotspot, using transformer encoders to enhance feature extraction capabilities and improve feature matching accuracy [20–26], particularly in handling images from different viewpoints, where they demonstrate a stronger ability to capture global information.

Observing the exceptional global context modeling ability and instance discrimination capability of transformer-based methods in visual tasks, we propose a saliency-aware integrated global-local network (SIGN) to effectively address the viewpoint differences and scale inconsistencies between UAV and satellite images. Specifically, SIGN employs a transformer-based DINOv2 [27] network for fine-grained feature extraction. Through the Saliency-Aware Partition (SAP) module, images are divided into multiple key regions. Our model integrates global feature contrast learning with saliency-aware local block contrast learning, enabling precise alignment at both global and local levels. Furthermore, classification is performed on both global features and saliency-aware local block features, enhancing the distinction between different image regions while addressing the challenges of viewpoint vari-

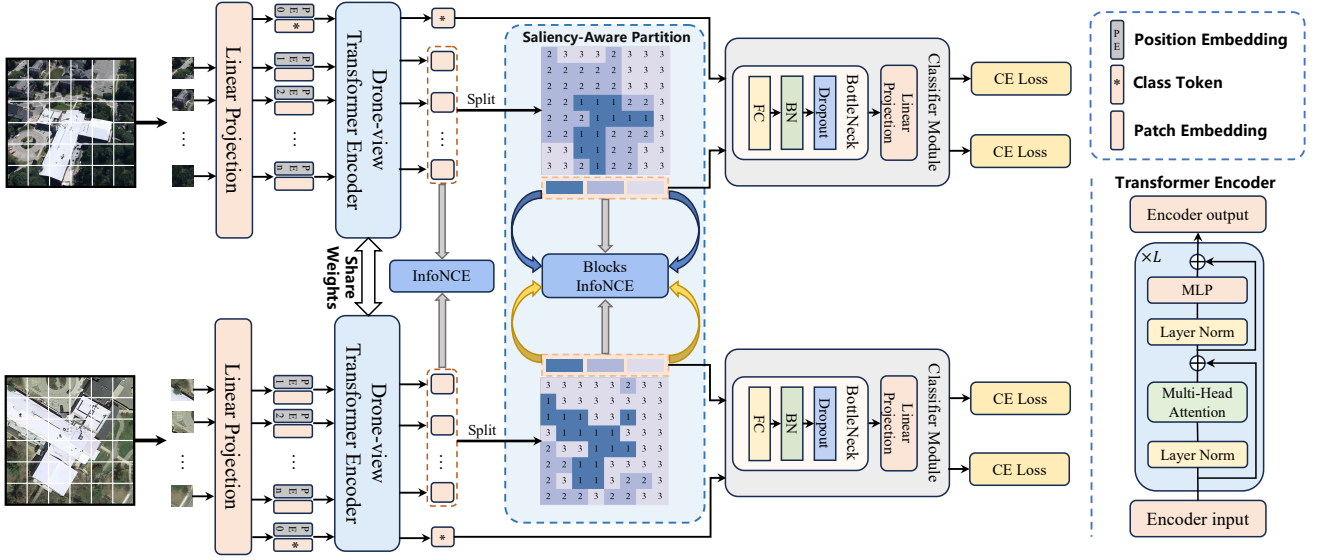


Fig. 1. Overview of the proposed SIGN model. SIGN consists of three main components: a transformer-based feature extraction network, the SAP module, and a classification module. It processes paired UAV and satellite images, extracts global features, divides images into key regions for local alignment, and classifies both global and local features to enhance feature matching precision.

ations and scale discrepancies. This multi-level optimization strategy significantly improves the accuracy and robustness of cross-platform localization.

Overall, the main contributions of this paper can be summarized as follows:

- We propose a novel CVGL method called SIGN, which employs DINOv2 network for fine-grained feature extraction and introduces the SAP module to effectively address the challenges of viewpoint and scale variations.
- The SAP module divides images into multiple key regions, enabling the integration of global feature contrast learning with saliency-aware local block contrast learning. This approach applies local contrastive loss for fine-grained alignment and performs classification on both global and local features to enhance region distinction and alignment accuracy.
- Extensive experiments on the University-1652 dataset demonstrate that SIGN achieves state-of-the-art (SOTA) performance, exhibiting superior accuracy and robustness in handling viewpoint variations and scale discrepancies.

II. METHOD

A. Overall Architecture

The overall architecture of the proposed SIGN is illustrated in the Fig. 1. The model consists of three main components: the transformer-based feature extraction network, the SAP module, and the classification module. SIGN consists of two branches, these two branches process two data streams simultaneously while sharing the same network weights. The input to SIGN consists of paired 518×518 pixel images, one from the UAV viewpoint and the other from the satellite viewpoint. Initially, a transformer encoder is employed to extract global features from both images, with symmetric InfoNCE serving as the contrastive learning loss function. Next, we introduce

the SAP module, which partitions the global feature map into multiple saliency-aware regions and performs matching on these regions, thereby improving feature alignment accuracy. Finally, both global and local features are processed through a classifier to compute their respective class probabilities. The classifier minimizes the discrepancy between predicted and true class probabilities using a cross-entropy loss function, thereby enhancing the utilization of semantic information and facilitating more accurate cross-view localization.

B. Feature Extraction Process

To effectively exploit the representation information from both UAV and satellite perspectives and enhance cross-view alignment, we employ a weight-sharing Siamese network architecture. The network processes UAV and satellite images in parallel, leveraging the two viewpoints to jointly optimize the network parameters, thereby achieving efficient fusion and alignment of multi-view features. To address performance bottlenecks in feature extraction, we introduce DINOv2 as the backbone network for feature extraction, aiming to further enhance the robustness and generalization ability of the feature representations. DINOv2 is a self-supervised learning-based vision transformer model that extracts task-agnostic, general visual features through unsupervised training on large-scale datasets, demonstrating powerful feature modeling capabilities. This backbone design effectively captures rich semantic information from input images, providing a robust foundation for cross-view feature matching and localization.

In this study, we fine-tune the DINOv2-L model with a detailed strategy. During the fine-tuning process, we freeze the parameters of the first 22 layers of the DINOv2-L network, allowing them to participate only in the forward pass without gradient computation, thereby reducing computational overhead and stabilizing the training process. Only the last two layers of the network are fine-tuned, where both forward

pass and gradient computation are performed for parameter updates. Freezing the lower layers preserves the original feature extraction capability, while the higher layers are trained to adapt to the specific UAV-satellite image matching task. Input images are first processed by a Patch Embedding module with a kernel size of 14 and a stride of 14, which divides the input images into 14×14 patches and maps them into the appropriate embedding space. The processed features are then passed into the DINOv2 model for further feature extraction. The output feature map has a size of $\frac{H}{14} \times \frac{W}{14}$, with 1024 feature channels.

C. Saliency-Aware Partition

To effectively address the scale variations and viewpoint discrepancies in CVGL tasks, traditional global contrastive learning methods, while improving feature alignment to some extent, remain inadequate in fully capturing the local differences present in UAV and satellite image matching. To overcome this limitation, our study introduces the SAP module. By employing a saliency-aware mechanism, the SAP module divides the image into local patches, extracting and aligning key regional features to enhance the precision of capturing detailed information between images. Specifically, the SAP module utilizes saliency guidance to segment the image into multiple critical regions, which are typically associated with the key features of the target object or scene. The saliency-aware partitioning process effectively focuses on the most informative areas of the image, providing finer and more meaningful representations for cross-view image matching, thus improving local feature alignment. This method captures the similarities in key regions of images from different perspectives, thereby enhancing the precision of image matching at a finer-grained level.

Contrastive learning is a widely used technique in CVGL tasks, whose core idea is to optimize feature space alignment by constructing positive and negative sample pairs. To ensure the model captures useful features effectively, large batch sizes are usually required to construct a rich positive-negative sample dictionary. However, increasing batch sizes introduces substantial computational and memory overhead, particularly when processing high-resolution images, thereby rendering this approach impractical in many scenarios. Therefore, improving contrastive learning performance without adding to the computational burden becomes a crucial challenge.

To address the challenge of local feature alignment, our study introduces a novel loss function termed Blocks InfoNCE. Specifically, the saliency-aware local features generated by the SAP module are used to compute local contrastive loss, thereby further enhancing the precision of local feature alignment across images. Compared to global contrastive learning methods, local contrastive learning allows for more detailed capture of critical regions in the image, thereby increasing the robustness of image matching. To ensure a comprehensive alignment, we combine the global contrastive loss, which captures holistic contextual relationships, with the local contrastive loss, focusing on fine-grained features within salient

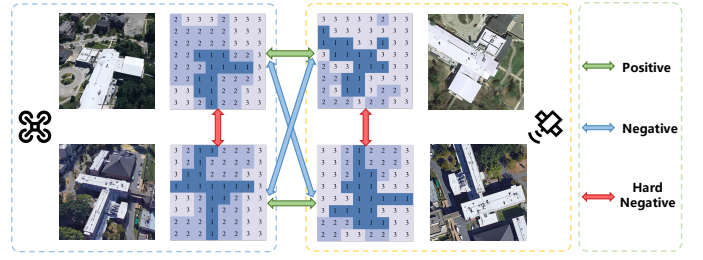


Fig. 2. By leveraging relationships between images from the same platform, BlocksInfoNCE introduces a substantial number of additional negative samples, termed hard negatives, without incurring extra memory overhead.

regions. Moreover, to improve the distinguishing capability of the feature extractor across different scenes, we incorporate contrastive constraints between images from the same platform. By increasing the number of negative samples, we effectively expand the sample space for contrastive learning, enhancing the model's sensitivity to visual information. These negative samples, referred to as hard negatives, originate from different scenes within the same platform and present greater challenges. As illustrated in Fig. 2, the BlocksInfoNCE loss leverages these hard negatives by incorporating relationships between images from the same platform, such as Drone-to-Drone and Satellite-to-Satellite comparisons. This approach allows for an increased variety of negative samples without additional memory costs, improving the robustness of feature alignment across diverse viewpoints. By introducing hard negatives, the model not only benefits from more negative sample signals but also improves its discriminative ability during optimization, further enhancing feature alignment across images. The computation of the loss can be expressed using the following equations:

$$\mathcal{L}_{\text{InfoNCE}}(q, R) = -\log \frac{\exp(q \cdot r_+ / \tau)}{\sum_{i=0}^R \exp(q \cdot r_i / \tau)}, \quad (1)$$

$$\mathcal{L}_{\text{BlocksInfoNCE}}(q, R) = -\frac{1}{N} \sum_{\mathcal{P}} \sum_{j=1}^N \log \frac{\exp(q_j^{\mathcal{P}} \cdot r_{j+}^{\mathcal{P}} / \tau)}{\sum_{i=1}^R \exp(q_j^{\mathcal{P}} \cdot r_{j,i}^{\mathcal{P}} / \tau)}. \quad (2)$$

The InfoNCE loss function, as shown in Eq. (1), measures the similarity between the query feature q and a set of reference features R by using the dot product. The positive sample uniquely corresponding to the query feature q is denoted as r_+ . The InfoNCE loss is lower when the query and the positive match are similar, and higher when the negative sample r_i is dissimilar to the query q . The BlocksInfoNCE loss, as defined in Eq. (2), extends the InfoNCE formulation to patch-level features. In this context, the set $\mathcal{P}$ represents different matching scenarios incorporated into the loss computation, including Drone-to-Satellite (D-S), Drone-to-Drone (D-D), and Satellite-to-Satellite (S-S) comparisons. The term $q_j^{\mathcal{P}}$ represents the feature of the j -th block in the query feature q , while $r_{j+}^{\mathcal{P}}$ denotes the uniquely similar positive block feature corresponding to $q_j^{\mathcal{P}}$ within the reference feature R . The set

of negative features, $r_{j,i}^{\mathcal{P}}$, includes non-matching blocks within the reference set R . The parameter τ is a temperature scaling factor that adjusts the sharpness of the similarity distribution.

In CVGL tasks, each target can be treated as an individual class, allowing the concept of image classification to be effectively integrated into CVGL to enhance the precision and reliability of target localization. Based on this idea, we input both global features and saliency-aware local features into the classification module, employing a cross-entropy loss function to measure the discrepancy between the true and predicted probability distributions. By minimizing this loss, the network continuously optimizes its parameters during training, thus improving the accuracy and robustness of cross-view feature matching. The classification loss for the global and saliency-aware local features can be computed using the following formula:

$$\mathcal{L}_{\text{CE}}^{\text{global}}(F_{\text{global}}, \text{label}) = -\log \frac{\exp(F_{\text{global}}[\text{label}])}{\sum_j \exp(F_{\text{global}}[j])}, \quad (3)$$

$$\mathcal{L}_{\text{CE}}^{\text{local}}(F_{\text{local}}, \text{label}) = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(F_{\text{local},i}[\text{label}])}{\sum_j \exp(F_{\text{local},i}[j])}, \quad (4)$$

$$\mathcal{L}_{\text{CE}} = \mathcal{L}_{\text{CE}}^{\text{global}} + \mathcal{L}_{\text{CE}}^{\text{local}}, \quad (5)$$

where F_{global} and F_{local} represent the global and local feature maps extracted from drone or satellite images, while label represents the geographic region label corresponding to the image, and the set j contains all the scene labels.

This classification strategy not only strengthens the model's discriminative power but also facilitates the alignment of target regions across different viewpoints, further improving image matching precision.

The total loss combines global and local contrastive losses with classification loss, balanced by weighting factors, to jointly optimize feature alignment and classification accuracy.

$$\mathcal{L}_{\text{loss}} = \mathcal{L}_{\text{InfoNCE}} + \mathcal{L}_{\text{BlocksInfoNCE}} + \mathcal{L}_{\text{CE}}. \quad (6)$$

III. EXPERIMENTS

A. Dataset

This paper conducts experiments on the University-1652 [8] dataset, which is a large-scale UAV-satellite cross-view geolocalization dataset. The dataset integrates data from three platforms: drones, satellites, and ground-level imagery, and includes 1,652 buildings across 72 universities worldwide. It contains 50,218 training images, 41,135 query images, and 55,227 gallery images, supporting two critical tasks: drone-view target localization (Drone $\rightarrow$ Satellite) and drone navigation (Satellite $\rightarrow$ Drone).

B. Implementation Details

The experiments were conducted on an NVIDIA 48GB A40 GPU, using the PyTorch deep learning framework. The input images were resized to 518×518 pixels, and the batch size was set to 64. The entire training process was completed in a single epoch, employing the AdamW optimizer with an initial learning rate of 0.0005. A cosine learning rate scheduler was utilized over the course of this epoch, with the first 10% designated as the warm-up phase.

C. Results

The performance of the proposed SIGN model was evaluated on the University-1652 dataset, and the results are summarized in Table I. SIGN achieved the best performance in both the Drone $\rightarrow$ Satellite and Satellite $\rightarrow$ Drone tasks, demonstrating superior accuracy and robustness in handling viewpoint variations. The results indicate that the proposed method effectively addresses the challenges of scale discrepancies and viewpoint differences in CVGL tasks, significantly improving feature alignment and localization accuracy.

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE UNIVERSITY-1652 DATASET.

Method	Drone2Sat		Sat2Drone	
	R@1	AP	R@1	AP
LPN [14]	75.93	79.14	86.45	74.79
FSRA [20]	82.25	84.82	87.87	81.53
MBF [21]	89.05	90.61	93.15	88.17
MCCG [15]	89.64	91.32	94.30	89.39
Sample4Geo [16]	92.65	93.81	95.14	91.39
TransFG [22]	84.01	86.31	90.16	84.61
GeoFormer [23]	89.08	90.83	92.30	88.54
SeGCN [24]	89.18	90.89	94.29	89.65
MFJR [25]	91.87	93.15	95.29	91.51
SRLN [26]	92.70	93.77	95.14	91.97
CAMP [17]	94.46	95.38	96.15	92.72
SIGN(Ours)	94.68	95.59	96.29	94.33

D. Ablation Study

Effectiveness of Different Backbone Models. To evaluate the impact of different backbone models on the performance of SIGN, we conducted ablation studies using different sizes of the DINOv2 model. The results are summarized in Table II.

TABLE II
ABLATION STUDY ON DIFFERENT BACKBONE MODELS.

Backbone	Drone2Sat		Sat2Drone	
	R@1	AP	R@1	AP
DINOv2-Small	91.45	92.77	93.86	91.40
DINOv2-Base	93.68	94.72	95.32	93.87
DINOv2-Large	94.68	95.59	96.29	94.33

The results indicate that the performance of SIGN improves with the size of the DINOv2 model. The DINOv2-L backbone achieves the best results, demonstrating the importance of using a powerful feature extraction network for CVGL tasks.

Effectiveness of SAP Module and Classification Module.

To evaluate the impact of the SAP module and the classification module on the performance of SIGN, we conducted ablation studies by adding these components to the baseline one at a time. The results are summarized in Table III.

TABLE III
THE RESULTS OF ABLATION STUDIES OF PROPOSED METHOD ON UNIVERSITY-1652.

Method	Drone→Satellite		Satellite→Drone	
	R@1	AP	R@1	AP
Baseline	93.70	94.70	94.56	90.90
w/ SAP	94.46 ↑ 0.76	95.36 ↑ 0.66	95.65 ↑ 1.09	92.89 ↑ 1.99
w/ Classification	94.42 ↑ 0.72	95.34 ↑ 0.64	95.72 ↑ 1.16	93.58 ↑ 2.78
SIGN (Ours)	94.68 ↑ 0.98	95.59 ↑ 0.89	96.29 ↑ 1.73	94.33 ↑ 3.43

The results indicate that both the SAP module and the classification module contribute to the performance of SIGN in CVGL tasks. The SAP module enhances the model’s ability to capture and align local features, addressing the challenges of scale variations and viewpoint discrepancies. The classification module further improves the model’s discriminative power by leveraging semantic information, facilitating more accurate cross-view localization. Removing either module results in a noticeable drop in performance, demonstrating the importance of these components in achieving sota results in CVGL tasks.

E. Conclusion

In this paper, we propose a novel CVGL method named SIGN, which effectively addresses the challenges of viewpoint differences and scale inconsistencies between UAV and satellite images. SIGN employs a transformer-based DINOv2 network for fine-grained feature extraction and introduces the SAP module to divide images into key regions for local alignment. The model integrates global feature contrast learning with saliency-aware local block contrast learning, enhancing alignment accuracy at both global and local levels. Experiments on the University-1652 dataset demonstrate that SIGN achieves sota performance, demonstrating superior accuracy and robustness in handling viewpoint variations. In future work, we plan to explore the application of SIGN in real-world scenarios and further investigate the potential of transformer-based methods in cross-view image matching.

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